

Silent Failure in Semi-Supervised Sound Event Detection: Synthetic Validation Can Hide Real-World Collapse

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Abstract—Semi-supervised sound event detection models are routinely trained on synthetic strongly labeled, real weakly labeled, and real unlabeled audio, and selected on validation data that contains no strongly labeled real recordings. We show that this practice can fail silently. Under the classical DCASE protocol, Mean Teacher can degrade and even collapse to near-zero PSDS1 on real recordings while both quantities used for selection, the validation scores on the synthetic data and on a weakly labeled real data subset, remain unremarkable. Across nine architecture configurations we identify two geometrically distinct failure modes: sharp minima with elevated real-domain curvature, and flat plateaus of elevated real-domain loss. No quantity we examined that is computable without frame-level real labels separates failing from non-degraded runs. Among the diagnostics we test, the cheapest that separates non-degraded, degraded, and collapsed runs is the ratio of supervised loss on real versus synthetic audio, computed on held-out real evaluation data with frame-wise labels and therefore available only post hoc; its frame-level component reveals that what silently breaks is temporal localization. A seed-paired supervised-only control shows the failure is not specific to Mean Teacher; its objective widens a supervised instability in the sharp mode, while the flat mode did not occur without it. Replacing Mean Teacher with FlatMatch avoids both failure modes. On a 3M-parameter MobileNet where Mean Teacher reaches at most 0.33 PSDS1 and four of twelve seeds collapse to zero, FlatMatch reaches 0.46, within 0.04 of a 30-times larger CRNN+BEATs baseline. FlatMatch outperforms Mean Teacher on the recurrence-free MobileNet configurations and the transformer, while Mean Teacher stays ahead on every recurrent configuration; flatness-aware training acts as insurance against silent failure rather than as a universal performance booster.

Index Terms—sound event detection, semi-supervised learning, domain shift, loss landscape, model selection

1. INTRODUCTION

Sound event detection (SED) requires frame-level annotations that are scarce and expensive, while unlabeled in-domain audio is abundant. Semi-supervised learning is therefore central to the classical DCASE Task 4 setting (2019–2023), which builds on a strongly labeled (frame-level) synthetic set together with weakly labeled (clip-level) and unlabeled real recordings [1], [2]. The Mean Teacher (MT) framework [3] has been the semi-supervised component of the Task 4 baselines [2], [4], and teacher-based consistency remains at the core of current systems [5]–[7]. Under this protocol, the only strong labels for training come from synthetic audio, and models are selected on a synthetic validation set complemented by a small weakly labeled real subset, which implicitly serves as a proxy for real-audio performance.

We show that this proxy can fail silently. Figure 1 previews it on a 3M-parameter frame-wise MobileNet [8] pretrained on AudioSet strong [9] (fmn10_pre), comparing MT with FlatMatch (FM) [10], the flatness-based remedy we examine.

MT remains the SED default even though newer semi-supervised methods have surpassed it in single-label classification [11]–[13]. Most rely on argmax pseudo-labels and do not transfer cleanly to multi-label SED. FM is an exception. Its consistency criterion does not require such pseudo-labels and leaves the pipeline unchanged. Unlike MT, it explicitly shapes the loss geometry (Section 2.2), where our diagnosis locates one of the two failure modes.

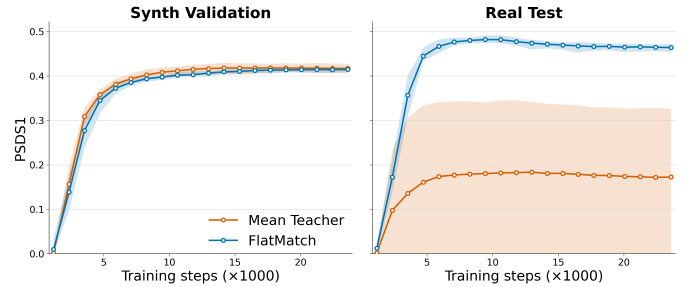


Fig. 1: Synthetic validation (left) and real-domain PSDS1 [14] (right) for the pretrained MobileNet `fmn10_pre`, mean and min–max envelope across twelve seeds per method. The validation curves are indistinguishable; on real audio four MT seeds collapse to zero. It is the only configuration whose MT outcomes span all three zones (Table 1).

This paper makes four contributions. First, we document the failure and establish its silence in a study spanning nine architecture configurations: whenever MT degrades or collapses on real audio, every quantity computed during standard training (Section 3) remains unremarkable, on synthetic data, weakly labeled real clips, and unlabeled real recordings alike. Second, we diagnose the failure geometrically. It occurs in two distinct modes — sharp minima with strongly elevated directional curvature on real audio, and flat plateaus of elevated real-domain loss — which share a single post-hoc signature; in every failing run the supervised loss on the held-out real evaluation data rises above its synthetic counterpart, driven by its frame-level component, that is, by temporal localization. Third, a seed-paired supervised-only control attributes the failure. Half of the sharp-mode collapses persist without the MT term, equally silently, and the term widens a failure basin already present in the supervised case; the flat-plateau mode did not occur without it.

Fourth, we show that replacing MT with FM avoids both failure modes, and that the gain is not merely the removal of the MT term; on `fmn10_pre` a paired ablation ties it to FM’s own semi-supervised term. We characterize the scope of this fix, whose advantage does not extend to recurrent architectures.

2. BACKGROUND

2.1. Mean Teacher

The MT framework [3], the semi-supervised objective of the Task 4 baseline, maintains a teacher network whose weights are an exponential moving average (EMA) of the student weights. The student is trained with a consistency loss that penalizes prediction disagreement between student and teacher on perturbed inputs. The consistency term requires no labels; it is the mechanism through which the unlabeled real recordings contribute to training.

2.2. Sharpness-aware learning and FlatMatch

A separate line of work links generalization to the flatness of the loss landscape: among solutions with comparable training loss, those

in broad, flat minima tend to generalize better than those in sharp minima [15]. The claim is conditional on the loss being comparably low; flatness of a poor solution carries no such benefit, a distinction that becomes relevant in Section 3. Sharpness-Aware Minimization minimizes the loss under a worst-case weight perturbation within a small neighborhood, explicitly favoring flat regions. FM [10] extends this idea to the semi-supervised setting: the worst-case perturbation is estimated from the labeled loss, and the model is then penalized for prediction inconsistency on unlabeled data between original and perturbed weights. In contrast to MT, consistency is thus enforced not between two networks but between a model and a perturbed copy of its own weights, a criterion that requires no argmax pseudo-labels and applies unchanged to multi-label SED. We use FM as a drop-in replacement for MT, changing only the semi-supervised objective while keeping architecture, data, and optimization otherwise fixed.

3. SILENT FAILURE DIAGNOSIS

This section defines what we measure and how we group runs; Section 5 reports what the measurements show. All diagnostics are evaluated separately on the synthetic validation data and on the labeled real recordings (Section 4), at each run’s final training checkpoint.

We group runs by real-domain PSDS1 before examining any diagnostic, with cutoffs read off our runs rather than calibrated. A run is collapsed if its real-domain PSDS1 falls below 0.06, degraded if it falls clearly below its configuration’s healthy seeds, separated by a gap exceeding their spread (Section 5.2), and non-degraded otherwise; Table 1 lists each configuration’s zone composition.

The first diagnostic is the supervised loss L , the average of frame-level and clip-level binary cross-entropy. As a per-run scalar summary we use the ratio of the real to the synthetic supervised loss. Under the standard protocol, frame-level labels for real audio do not exist at training time, so the numerator uses held-out real evaluation data; the ratio is by construction a post-hoc diagnostic, not a monitoring signal. We further decompose it into its frame-level and clip-level components to locate which form of supervision separates failing from non-degraded runs.

The second diagnostic is a directional curvature following the adaptive scaling of ASAM [16]. With w the model weights, g the gradient of L at w over the full set, and $T_w = \text{diag}(|w| + \eta)$ the elementwise scaling that measures each parameter’s perturbation relative to its own magnitude, we perturb along the adaptively normalized gradient direction and take a symmetric second difference,

$$\varepsilon^* = \rho \frac{T_w^2 g}{\|T_w g\|}, \quad \kappa = \frac{1}{2} [L(w + \varepsilon^*) + L(w - \varepsilon^*)] - L(w), \quad (1)$$

with $\rho = 0.05$ and $\eta = 10^{-2}$. The symmetric difference cancels the first-order term exactly, so $\kappa \approx \frac{1}{2} \varepsilon^{*\top} H \varepsilon^*$ measures curvature along this direction even away from a stationary point; a collapsed model need not sit at a stationary point of the real-domain loss, where κ is then a directional curvature rather than the curvature of a minimum. The perturbation has fixed length ρ in the metric induced by T_w , so κ is comparable across runs in that metric.

These two measurements will separate the failing runs into two geometric profiles, which we call the sharp and the flat mode as shorthand: sharp minima, with strongly elevated directional curvature on real audio at unremarkable synthetic curvature, and flat plateaus, with real-domain curvature comparable to non-degraded runs but elevated real-domain loss. The existence of a flat mode matters for the design of any diagnostic. A model can fail while remaining flat, so we build the ordering diagnostic on the loss rather than on curvature.

To establish that the failure is silent, we test every quantity the standard training protocol computes against the collapse/degraded/non-degraded grouping: the synthetic validation metrics used for selection (PSDS1, intersection-F1, and event-F1), the weak-real macro F1, the synthetic supervised loss, the weakly supervised loss on the real clips on the training set and the held-out validation subset alike, and the consistency loss on the unlabeled real recordings. Beyond the protocol, we also test the synthetic- and real-domain curvatures and raw Hessian-based sharpness measures.

Finally, we test whether the consistency loss is responsible for the failure. We repeat MT runs with the semi-supervised loss weight set to zero, keeping hyperparameters, batch composition, and seeds identical. The unlabeled and weakly labeled real clips remain in every batch and pass through the same forward computation, so batch-normalization statistics are still estimated from real audio. Only the consistency gradient is removed. Control seeds are selected by outcome rather than at random. For the one configuration whose outcome varies across seeds, we cover all four collapsed seeds, one degraded seed, and the best non-degraded seed. For the two configurations on which MT fails uniformly, we run one seed each. A second condition sets $\rho = 0$ under FM’s own settings, where the perturbation and with it the entire consistency term vanish identically, giving supervised training at FM’s hyperparameters for one seed.

4. EXPERIMENTAL SETUP

4.1. Data

All experiments follow the classical DCASE Task 4 protocol on DESED [1]. The strongly labeled training material is synthetic, mixing foreground events with backgrounds via Scaper [17], whereas the target recordings are real domestic audio. Training combines synthetic strongly labeled clips with 1421 weakly labeled and 14412 unlabeled real clips; selection uses only synthetic strongly labeled validation clips and the remaining 157 weakly labeled real clips. The synthetic train and validation splits share their background pool and split foreground events below the source-recording level, so high synthetic validation reflects within-distribution generalization, not transfer to real audio. Strongly labeled real audio exists only in the 1168-clip set reserved for evaluation, so frame-level real labels are unavailable during training by design. Later DESED editions add strongly labeled real training data [2], which overlaps with the validation set [18]; we deliberately study the classical composition, in use through DCASE 2023 and still adopted when strong real annotation is unavailable — exactly the regime our question concerns.

4.2. Models and configurations

We build on the DCASE 2023 Task 4 baseline pipeline [4] and vary the backbone architecture, its initialization, and the semi-supervised objective. The architectures span a 7-layer CNN (0.6M parameters), the CRNN baseline of DCASE 2023 Task4 a) [19] (the same CNN followed by a bidirectional GRU, 1.1M), and frame-wise MobileNet backbones [8] at two widths (fnn04, 0.5M; fnn10, 3.0M), the smaller one also with a GRU head (1.2M). The MobileNets are trained from scratch or from supervised AudioSet strong pretraining [8], [9], yielding eight configurations; a ninth is an 85.5M-parameter frame-wise audio spectrogram transformer (ASiT [20]) pretrained on AudioSet strong [21]. With an equal budget for both methods, we tune the learning rate, the EMA decay for MT, and the perturbation radius ρ for FM, selecting on synthetic validation PSDS1; the single exception is the CRNN under MT, which keeps its published settings. Following the FM authors, all tuned configurations use cosine learning-rate decay for both methods. All other pipeline components follow the baseline.

Table 1: Real-domain PSDS1 (1168 real clips, *sed_scores*; mean over seeds with min–max in brackets), zone composition (N = non-degraded, D = degraded, C = collapsed, Section 3), and the combined supervised loss ratio $L_{\text{real}}/L_{\text{synth}}$ (min–max). The *_pre* suffix denotes AudioSet strong pretrained initialization, otherwise from scratch. Three seeds per configuration except *fmn10_pre*, which uses twelve. Every FM run is non-degraded. Rules separate recurrence-free from recurrent configurations; FM leads on five of the six recurrence-free configurations, MT on the CNN and all recurrent ones. Best PSDS1 per row in bold.

Configuration	Parameters (M)	Mean Teacher			FlatMatch	
		PSDS1	Zones	L_r/L_s	PSDS1	L_r/L_s
CNN	0.6	0.191 [0.185, 0.194]	3N	0.58–0.60	0.180 [0.174, 0.186]	0.50–0.56
<i>fmn04_pre</i>	0.5	0.337 [0.319, 0.369]	3N	0.66–0.71	0.375 [0.360, 0.387]	0.56–0.60
<i>fmn04</i>	0.5	0.000 [0.000, 0.001]	3C	1.23–1.33	0.214 [0.175, 0.240]	0.62–0.73
<i>fmn10_pre</i>	3.0	0.172 [0.000, 0.326]	6N/2D/4C	0.65–12.6	0.464 [0.453, 0.472]	0.50–0.54
<i>fmn10</i>	3.0	0.025 [0.000, 0.051]	3C	1.31–1.44	0.228 [0.207, 0.240]	0.69–0.78
<i>ASiT_pre</i>	85.5	0.452 [0.449, 0.453]	3N	0.42–0.46	0.466 [0.464, 0.468]	0.57–0.59
CRNN	1.1	0.359 [0.352, 0.363]	3N	0.66–0.69	0.329 [0.321, 0.338]	0.57–0.59
<i>fmn04_pre</i> +GRU	1.2	0.431 [0.423, 0.438]	3N	0.67–0.72	0.394 [0.378, 0.413]	0.68–0.72
<i>fmn04</i> +GRU	1.2	0.341 [0.339, 0.343]	3N	0.73–0.77	0.305 [0.304, 0.307]	0.65–0.67

4.3. Evaluation protocol

We report PSDS1 [14] in the *sed_scores* implementation [22], following the Task 4 scenario-1 configuration, which evaluates all distinct operating points jointly rather than a fixed finite set of thresholds. For MT we report the teacher, whose scores exceed the student’s marginally; FM and the supervised-only control are reported as the student. Real-domain scores are computed on the 1168 recordings of the DESED real *validation* set. As our central question is whether synthetic validation certifies real-domain reliability, we do not select a single configuration on synthetic validation and report its real score; we report real-domain performance for every configuration, method, and seed, so that the relationship between validation and transfer is visible rather than assumed. Some of these real recordings are drawn from AudioSet [23] and thus overlap with our backbones’ pretraining data. We report results on the full set and release our code, including a list of overlap-free clips (<https://github.com/Jonathan-Greif/silent-failure-sed>).

5. RESULTS

5.1. Collapse, attribution, and remedy

Every FM run is non-degraded. Table 1 reports real-domain PSDS1, zone composition, and the loss ratio for all nine configurations. On *fmn10_pre*, MT reaches at most 0.33 PSDS1 across twelve seeds, four collapsing to zero, whereas FM reaches 0.46 on average. On the selection metric the collapse is invisible. One collapsed seed ties a non-degraded one on synthetic validation PSDS1, and the seed ranked highest there reaches only 0.25 on real audio. For reference, the CRNN+BEATs [24] baseline reaches 0.50 [19] with roughly 30 times more parameters, placing FM within 0.04 of a far larger system. Under FM the 85.5M ASiT reaches the same level as *fmn10_pre* while its MT counterpart stays below, with disjoint seed ranges.

The failure is not specific to MT, and its attribution depends on the failure mode. On *fmn10_pre* the MT term is not necessary for the collapse. Of the four seeds that collapse under MT, two also collapse without it (PSDS1 0.00), one lands degraded at 0.084, and one recovers to the lower edge of the non-degraded range (0.25). The supervised collapses are silent in the same way as MT’s; both their synthetic validation scores and their weak-real validation metrics are unremarkable. At the same time the MT term never helps on the seeds we pair. The degraded MT seed recovers to 0.32, the best MT seed improves from 0.33 to 0.38, and the supervised run matches or exceeds its MT counterpart in all six pairs. The supervised recipe thus carries a seed-dependent instability of its own, spanning zero to 0.38 across the six seeds, and the MT objective widens it, from two collapses among the paired seeds to four, while lowering the non-collapsed outcomes. On the scratch-trained MobileNets the MT

term is necessary for the failure. Where MT collapses in all three seeds, the control reaches 0.24 (*fmn10*) and 0.19 (*fmn04*), the level of FM on the same configurations (0.23 and 0.21). Both statements rest on one seed per scratch configuration and six paired seeds, and support existence claims rather than rates.

On *fmn10_pre*, FM’s semi-supervised term adds to supervised training rather than only avoiding harm. On the scratch-trained configurations, where supervised training already reaches FM’s level, FM merely avoids reintroducing the harm. On *fmn10_pre* it is positive. FM’s weakest seed exceeds the best supervised outcome at either setting (0.453 against 0.408, the $\rho = 0$ run), FM remains non-degraded on all twelve seeds, including the two on which even supervised training collapses, and its semi-supervised term is causally active; setting $\rho = 0$, which reduces FM to supervised training at unchanged settings, costs 0.058 PSDS1 on the same seed, three times the FM seed spread. The ablation does not isolate flatness, since $\rho = 0$ removes both the weight perturbation and the consistency term it induces, and we claim no mechanism for the recovery.

Recurrence bounds the FM advantage. FM outperforms MT on all four recurrence-free MobileNet configurations and on the transformer, with disjoint seed ranges throughout except on *fmn04_pre*, where they overlap. MT stays ahead elsewhere, decisively on every recurrent configuration and marginally on the recurrence-free CNN, by 0.011 on average, ranges touching at one seed pair. Neither parameter count (FM wins from the 0.5M MobileNet to the 85.5M transformer, while the 0.6M CNN sits between without an FM advantage), initialization, nor attention (the transformer sides with the MobileNets) explains the pattern. Recurrence is the strongest predictor. Adding a GRU to *fmn04*, on which FM otherwise wins, reverses the ranking in both initializations. The CNN blurs the boundary without breaking it, and the mechanism remains open (Section 5.3).

5.2. Quantifying failure modes

Collapse takes two geometric forms. Figure 2 plots the loss ratio against real-domain PSDS1 for all our runs. The collapsed MT runs realize both modes anticipated in Section 3. In the sharp mode, across the four collapsed seeds of *fmn10_pre*, the real-domain directional curvature is strongly elevated relative to the synthetic curvature of the same run, while the synthetic curvature itself remains in the range of non-degraded runs; on this configuration the curvature ratio separates the zones with disjoint ranges, 0.5 to 1.4 for the six non-degraded seeds, 1.8 and 3.3 for the two degraded, and 14 to 19 for the four collapsed, consistent with a continuous transition from health into the sharp mode. In the flat mode the real-domain curvature matches non-degraded runs, but the real-domain loss is elevated, 0.33 to 0.37 against 0.08 to 0.20 for non-degraded runs. The two modes coincide

with pretraining: the sharp-mode failures occur on `fmn10_pre` and the flat-mode failures on the scratch-trained `fmn04` and `fmn10`, which fail uniformly in all three seeds.

The real-to-synthetic loss ratio exceeds one in every failing run and stays below one in every non-degraded run. Under both semi-supervised objectives, non-degraded ratios lie between 0.42 and 0.90, degraded runs at 1.07 and 1.14, and collapsed runs between 1.23 and 12.6. The inversion point, a ratio of one, separates non-degraded from failing runs identically for both failure modes and does not depend on the two degraded labels, our softest assignments; setting them aside, every collapsed run sits above 1.2 and every non-degraded run below 0.90. The supervised control preserves the ordering. Its non-degraded runs span 0.55 to 0.87 and its failing runs 1.25 to 3.3, so inversion again separates the zones at a threshold consistent with one. Inversion is therefore necessary for failure under every objective; the semi-supervised objectives place the separation exactly at one, and the supervised control is consistent with the same point rather than contradicting it. Relative to the FM run with the same seed, setting $\rho = 0$ leaves the ratio essentially unchanged (0.53 to 0.55) while costing 0.058 PSDS1, so at this seed the semi-supervised term improves transfer without shifting the loss ratio.

What breaks is temporal localization. Decomposing the ratio locates the separation. It is driven by the frame-level term; in failing runs the frame-level real loss reaches 2.1 to 35 times its synthetic counterpart, while the clip-level ratio stays below one in every failing run, and the collapsed `fmn10_pre` seeds even show clip-level real losses below those of healthy CNN runs. Clip-level supervision on the same held-out clips is thus blind to the failure. The failing models still recognize which events occur, but no longer when, and their predictions are not clip-constant. The frame-level ratio alone carries no global inversion point, as every non-degraded run on `fmn10_pre` also exceeds one there (1.2 to 1.9); averaging in the clip-level term normalizes it to one across configurations, hence the combined loss.

Without frame-level real labels, no quantity we examined detects the failure. This is the central negative result of this paper. It holds for the synthetic validation metrics (PSDS1, intersection-F1, event-F1), the synthetic supervised loss, and the synthetic curvature, whose failing values all lie inside the non-degraded range. It equally holds for the weakly supervised loss on the real clips, on the 1421 training clips and the 157-clip validation subset alike, for the weak-real macro F1, and for the consistency loss on the unlabeled recordings; on collapsed `fmn10_pre` runs student–teacher disagreement is of order 10^{-6} , so student and teacher agree, and only labels reveal that both are wrong. Even with real labels, curvature does not order the zones. A non-degraded CNN shows real-domain curvature up to $1.7 \times$ its synthetic while the flat-plateau failures show none, so curvature yields false positives and negatives. Hessian-based measures do not separate the zones either, consistent with their lack of reparameterization invariance and the inconsistent link between training-domain sharpness and generalization [25], [26].

5.3. Limitations

The loss-ratio signature requires frame-level real labels, so it characterizes the failure rather than providing a deployable monitor. Its practical content is negative; no label-free or weakly labeled quantity we examined detects the failure.

The ratio also orders zones, not ranks, and is blind to capacity underuse; on `fmn10_pre` even the best MT seed remains more than 0.13 PSDS1 below the FM mean at a ratio well inside the non-degraded range. The thresholds are read off our DESED runs rather

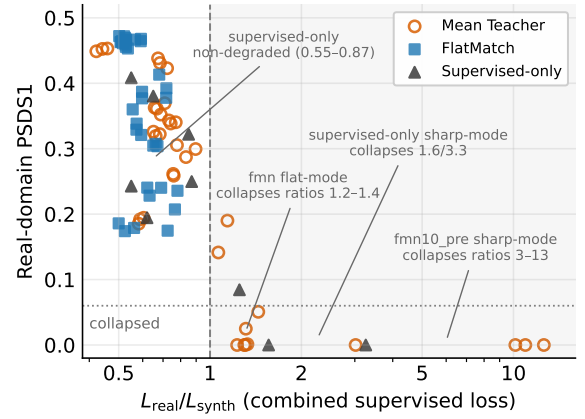


Fig. 2: Combined supervised loss ratio $L_{\text{real}}/L_{\text{synth}}$ versus real-domain PSDS1, one point per run: MT (teacher) and FM on all nine configurations plus the nine supervised-only control runs. Within every objective the ratio separates failing from non-degraded runs at an inversion point consistent with one (dashed): the two semi-supervised objectives place the separation exactly there, and the supervised-only runs (non-degraded 0.55–0.87, failing 1.25–3.3) are consistent with the same point. The dotted line marks the collapsed threshold; MT collapses split into flat-mode (1.23–1.44) and sharp mode (3–12.6) ratios, the supervised collapses (1.6, 3.3) fall in the sharp-mode only.

than calibrated, and the inversion point of one is an empirical value, not theoretically distinguished.

The supervised control covers three of the nine configurations, one seed each on the scratch-trained ones; with no control on the recurrent ones, MT’s advantage there is established relative to FM only.

Our geometric findings are observational and we did not intervene on sharpness directly. The coincidence of the two failure modes with pretraining rests on three seeds per configuration, twelve on `fmn10_pre`. Finally, the FM advantage does not extend to recurrent architectures, and the mechanism is unknown. Consistent with a task- and architecture-bound rather than universal benefit, an exploratory analysis in acoustic scene classification found sharper minima generalizing better, even under device shift [27].

6. CONCLUSION

The synthetic validation score and a small weakly labeled real subset are the only signals the classical DCASE Task 4 protocol uses for model selection, and both can certify models whose real-domain performance has silently collapsed. The failures take two geometric forms, yet the silence is uniform. No quantity we examined that is computable without frame-level real labels, on either domain, separates failing from non-degraded runs. The flat-plateau mode did not occur without the MT term, but half of the sharp-mode collapses persist without it, equally silently, and there the semi-supervised term widens an instability already present in the supervised objective. The one separating signal, the ordering of the supervised loss on real versus synthetic audio, is computed on the held-out evaluation data the protocol withholds from training. It characterizes the failure, orders the zones under every objective we test, and identifies what breaks as temporal localization. Replacing MT with FM avoids both modes and, on `fmn10_pre`, outperforms the supervised control, a margin its semi-supervised term carries. Since the advantage does not extend to recurrent architectures, flatness-aware training is insurance against silent failure, not a universal booster. Computing the worst-case perturbation from unlabeled target-domain data rather than the labeled synthetic loss is a natural next step the diagnosis suggests.

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